TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (CHEMISTRY)

CHAPTER 4: PERIODIC CLASSIFICATION

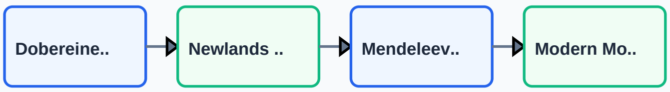
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Chemistry)
Chapter Name:	Periodic Classification

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: HISTORY OF PERIODIC CLASSIFICATION



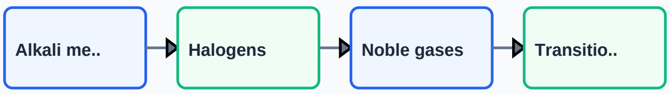
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Modern Periodic Law	The physical and chemical properties of elements are periodic functions of their atomic numbers.
Groups	The 18 vertical columns in the periodic table, where elements share the same number of valence electrons and chemical behaviors.
Periods	The 7 horizontal rows in the periodic table, representing the number of electron shells in the atom.
Atomic Radius	The distance from the center of the nucleus to the outermost valence shell of the electron cloud in an atom.
Electronegativity	The tendency of an atom in a molecule to attract the shared pair of electrons towards itself.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

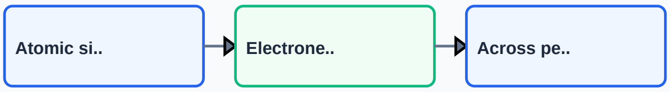
MODERN PERIODIC CLASSIFICATION GROUPS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: PERIODIC TRENDS ACROSS PERIODS



Essential Formulas, Laws & Identities:

- Atomic size trend: Decreases across a period (increasing nuclear charge), increases down a group (new shells added).
- Ionization Energy trend: Increases across a period, decreases down a group.
- Metallic character trend: Decreases across a period, increases down a group.

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Explain why Sodium (Z=11) is more reactive than Lithium (Z=3) although they belong to the same alkali metals group.

Step-by-step Solution:

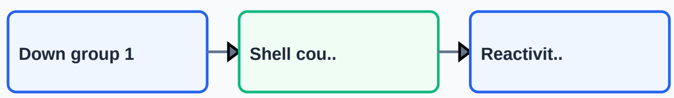
Sodium is lower in Group 1 than Lithium. It has more electron shells, making its valence electron further from the nucleus. Thus, less energy is required to lose this electron, making Sodium more reactive.

Example 2: Arrange the following in increasing order of atomic radius: F, N, O, B.

Step-by-step Solution:

These belong to the same Period 2. Atomic size decreases from left to right as nuclear charge increases. Order: B, C, N, O, F. So increasing order: F < O < N < B.

WORKED: EXPLAINING CHEMICAL REACTIVITY TRENDS



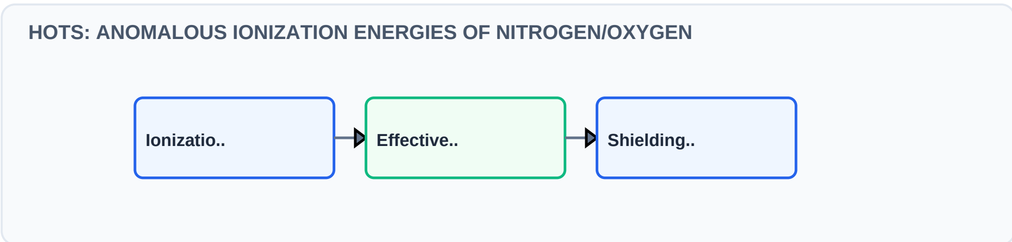
HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why are noble gases placed in a separate group (Group 18) at the far right of the periodic table?

Analytical Solution Strategy:

Noble gases have a completely filled outer valence shell (helium has 2, others have stable octet 8). This configuration makes them chemically inert and highly stable. Placing them in Group 18 reflects their shared zero valency and isolates them from highly reactive elements.



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Dobereiner's Law of Triads and Newlands' Law of Octaves.

Q2 (2-Mark): State the differences between Mendeleev's periodic table and the Modern periodic table.

Q3 (2-Mark): Define ionization energy and explain its trend across a period.

Q4 (5-Mark): Why does electronegativity decrease down a group?

Q5 (5-Mark): Name the first element of: (i) Alkali metals, (ii) Halogens, (iii) Noble gases.

Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Chemical Material Synthesis

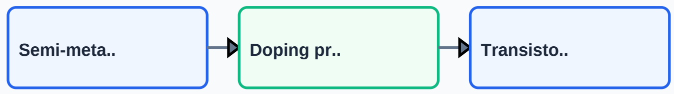
Engineers refer to group trends to predict material properties, designing alloys and semiconductors by matching elements.

Application Case: Alloy Manufacturing

Understanding atomic radii and periods allows metallurgists to replace iron atoms with chromium to form stainless steel.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: SEMICONDUCTOR FABRICATION PROPERTIES SELECTION



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: MODERN PERIODIC TRENDS OVERVIEW